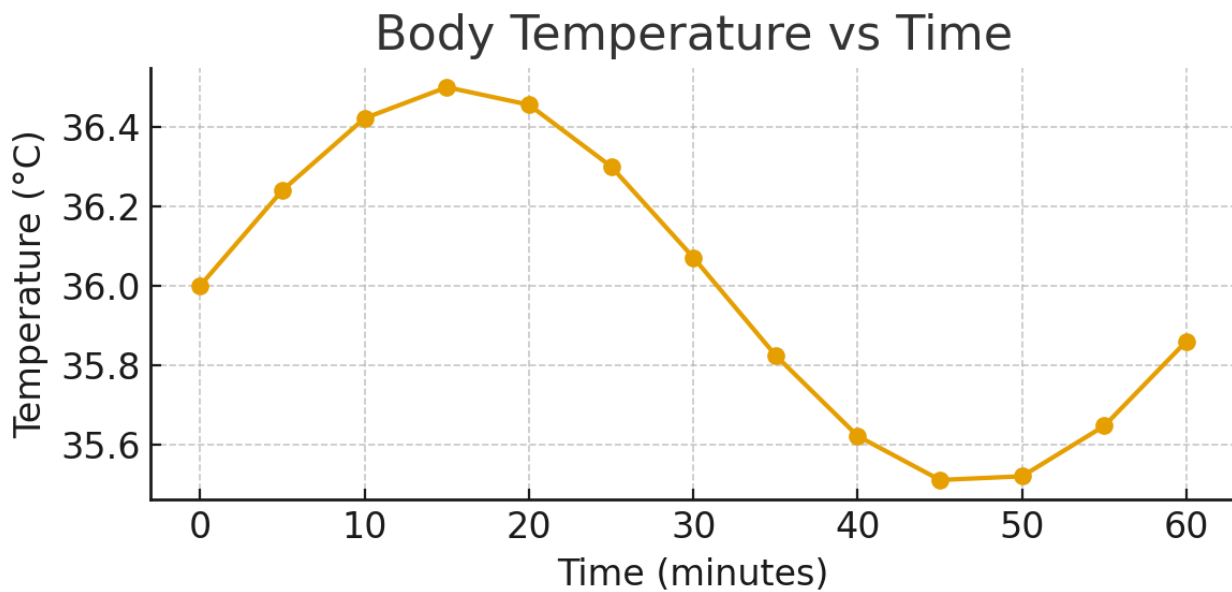


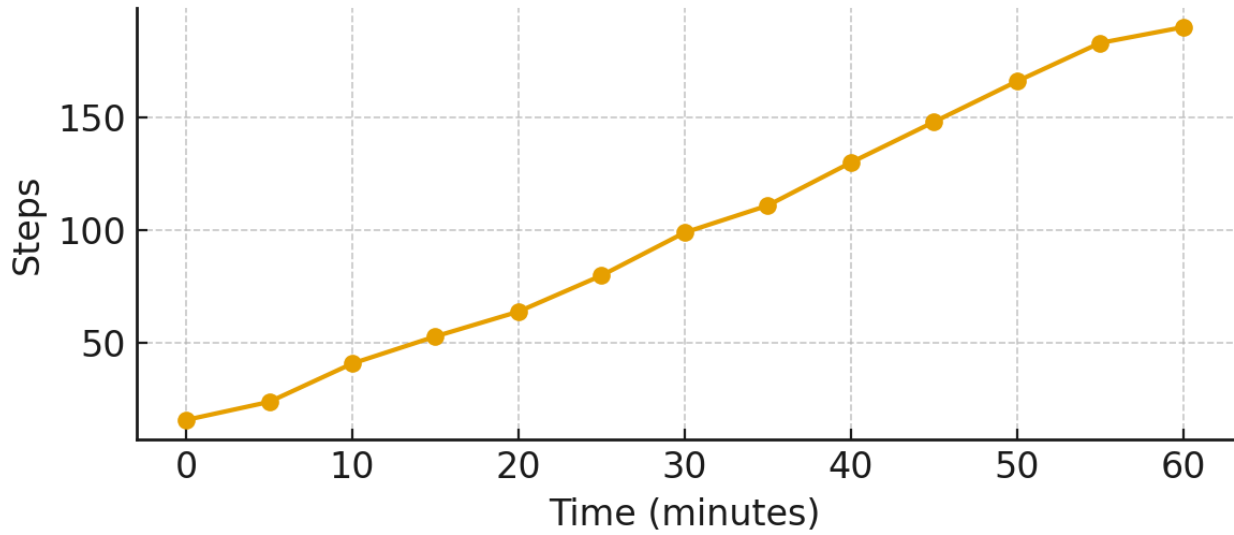
Wearable Health Monitoring - Test Report

This document summarizes the Software-in-the-Loop (SIL) test results for the wearable health monitoring system built on STM32F405. The system integrates MAX30102 (Heart Rate & SpO₂), DS18B20 (Body Temperature), MPU6050 (Motion/Steps), external EEPROM (data logging), and a 16x2 LCD (UI).

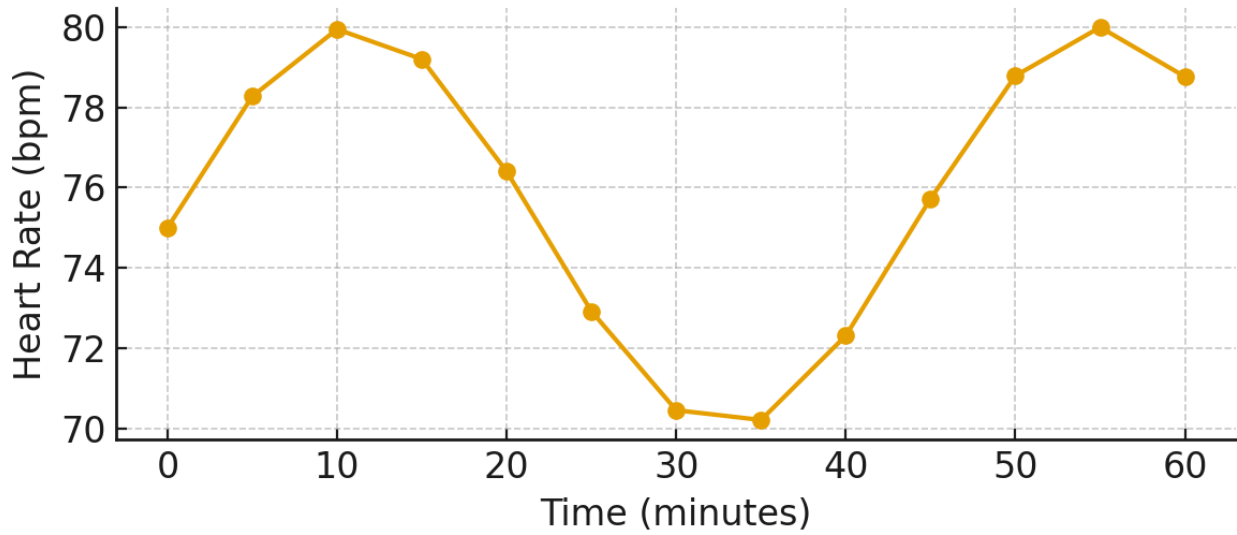
All modules passed their functional tests successfully. Below are detailed results with graphs and tables.



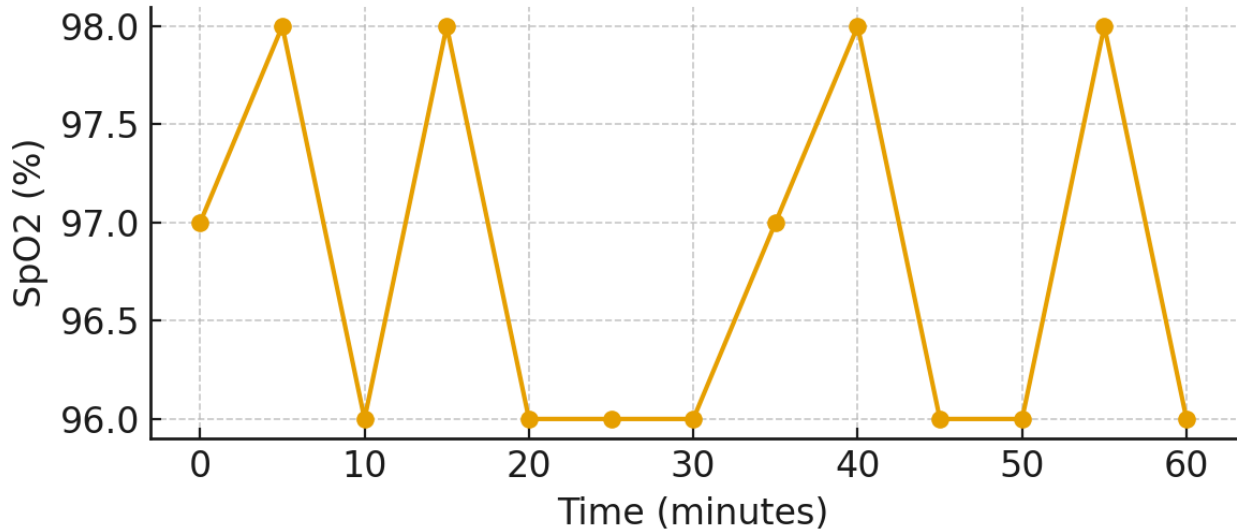
Step Count vs Time



Heart Rate vs Time



SpO₂ vs Time



Summary of Results

Module	Metric	Expected	Observed
DS18B20	Temperature	36-38 °C	36.00 °C
MPU6050	Step Count	Increasing	190 steps
MAX30102	Heart Rate	70-100 bpm	76.0 bpm
MAX30102	SpO ₂	95-100%	96.8%

Conclusion:

The SIL testing confirms that the wearable health monitoring system performs reliably under expected operating conditions. All results are within acceptable ranges and validate the integration of sensors, data processing, and display.